Trade Tariffs and Exchange Rates in a Multilateral World

Theoretical derivations

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1 Introduction

In a two-country model, an import tariff appreciates the tariffing country's currency under the Marshall–Lerner condition. In a multilateral setting, a discriminatory tariff also shifts expenditure across foreign suppliers. As a result, the tariffing country's bilateral exchange rates need not all move in the same direction.

These notes characterize the exchange-rate effects of tariffs in an N-country model. The general log-linearized equilibrium response is $(-J)^{-1}F$, where F is the vector of trade-balance disturbances created by the tariff at unchanged exchange rates and J is the Jacobian of trade balances with respect to exchange rates. Under symmetry and a multilateral Marshall–Lerner condition on J, the tariffing country always appreciates against the target, but its bilateral response against an untariffed bystander is ambiguous and depends on whether the cross-origin elasticity of substitution across foreign imported tradables exceeds some threshold.

Despite this bilateral ambiguity, the tariffing country's nominal effective exchange rate (NEER) appreciates for every admissible cross-origin elasticity and every N. We show that under symmetry the response to any tariff vector separates into an average tariff component and a relative tariff component, with the average tariff alone determining the effective exchange rate and tariff differentials determining the dispersion of bilateral responses around it (Section 5.2). Proofs are collected in Appendix E.

2 N-Country Model and Equilibrium

2.1 Production and preferences

Countries are indexed by $i, j \in \{1, \dots, N\}$. Where appropriate, we use A for the tariff imposer, B for the tariff target, and C for an untariffed bystander. Each country produces a country-specific tradable and a non-tradable from labor:

$$Y_{T_i} = A_{T_i} L_{T_i}, \qquad Y_{N_i} = A_{N_i} L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i,$$

with perfect competition and producer-price normalization $P_{T_i} = 1$ in i's currency, so

$$w_i = A_{T_i}, \qquad P_{N_i} = A_{T_i}/A_{N_i}.$$

Wages are constant in own currency, so relative adjustment occurs through exchange rates. Preferences are nested CES with share-linear weights:

$$\begin{aligned}
C_i &= C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} && \text{(tradable share } \alpha_T) \\
C_{T_i} &= \Big[\, \alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}} \Big]^{\frac{\eta}{\eta-1}} && \text{(home weight } \alpha_D,\ \text{elasticity } \eta) \\
M_i &= \Big[\, \textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}} \Big]^{\frac{\rho}{\rho-1}} && \text{(origin weights } \beta_{ij},\ \text{elasticity } \rho).
\end{aligned}$$

The elasticity η governs substitution between domestic and imported tradables, while ρ governs substitution across foreign origins conditional on importing. The latter margin is inactive at $N = 2$ and active at $N \geq 3$.

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N-1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Home bias α_D preserves an asymmetry between home and foreign tradables. Full symmetry between home and foreign tradables would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which we do not impose.

2.2 Tariffs, prices, income, and trade balance

Tariffs $\tau_{ij} \geq 0$ are ad valorem and are levied by i on imports from j, with $\tau_{ii} = 0$. Consumer prices in i's currency are $p_{ij} = (1+\tau_{ij})\, e_{ij}$ and $p_{ii} = 1$, that is, we normalize the price of the domestic tradable in its own currency to one. The corresponding price indices are

$$P_{M_i} = \Big[\textstyle\sum_{j\neq i} \beta_{ij}\, p_{ij}^{1-\rho}\Big]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \Big[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta}\Big]^{\frac{1}{1-\eta}}, \qquad P_i = \kappa_i\, P_{T_i}^{\alpha_T}\, P_{N_i}^{1-\alpha_T}. \tag{1}$$

Expenditure shares of income are

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}}\right)^{1-\eta}, \qquad s_{ij} = \alpha_T\, (1-\alpha_D) \left(\frac{P_{M_i}}{P_{T_i}}\right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}}\right)^{1-\rho}, \quad j \neq i, \tag{2}$$

so $s_{ij} I_i$ is tariff-inclusive spending at consumer prices and $s_{ij} I_i/(1+\tau_{ij})$ is the tariff-exclusive border value. Tariff revenue is rebated lump-sum:

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad \Longrightarrow \quad I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{3}$$

Equation (3) determines income in i's own currency. To add bilateral flows across countries, we express them in a common currency. Taking A's currency as the numeraire, define common-currency income as $\widetilde{I}_i \equiv e_{Ai}\, I_i$. Expenditure shares are ratios of same-currency values and are therefore unchanged by this conversion. Since the tariff wedge is a domestic transfer, trade balances are valued at producer prices in the common currency:

$$\mathrm{TB}_i = \sum_{k\neq i} f_{ki} - \sum_{j\neq i} f_{ij}, \qquad f_{ij} \equiv \frac{s_{ij}\,\widetilde{I}_i}{1+\tau_{ij}}, \tag{4}$$

where f_{ij} is the border value, in A's currency, of i's imports from j.

By Walras' law, only $N-1$ trade-balance conditions are independent. We also assume no internationally traded assets, so $\mathrm{TB}_i = 0$ for all i. Given $\{\tau_{ij}\}$, the $N-1$ free exchange rates therefore solve $\mathrm{TB}_i = 0$ for $i \neq A$.

2.3 Exchange-rate concepts

Define e_{ij} as units of i's currency per unit of j's, with $e_{ii} = 1$ and $e_{ij} = e_{ik}e_{kj}$. A rise in e_{ij} therefore represents a depreciation of i against j. Log changes in bilateral exchange rates relative to A are denoted by $\nu_j \equiv d\log e_{Aj}$, with $\nu_A = 0$. Relative wages, terms of trade, real exchange rates, and effective exchange rates are defined as

$$\omega_{ij} \equiv \frac{e_{ij}w_j}{w_i}, \qquad \mathrm{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij}P_{T_j}} = \frac{1}{e_{ij}}, \qquad q_{ij} \equiv \frac{e_{ij}P_j}{P_i}, \tag{5}$$

$$d\nu_i^E \equiv \sum_{j\neq i} w_{ij}\, d\log e_{ij}, \qquad dq_i^E \equiv \sum_{j\neq i} w_{ij}\, d\log q_{ij}, \qquad w_{ij} = \frac{f_{ij}+f_{ji}}{\sum_{k\neq i}(f_{ik}+f_{ki})}, \tag{6}$$

with $w_{ij} = 1/(N-1)$ in the symmetric model. Under this normalization of the domestic tradable's own-currency price,

$$d\log e_{ij} = d\log \omega_{ij} = -\,d\log \mathrm{ToT}_{ij}, \tag{7}$$

so the nominal exchange-rate results can equivalently be expressed in terms of relative wages or terms of trade.

3 Two Countries

The general linearization for N countries is given in Section 5.1; here we need only its $N = 2$ case. Write $m \equiv \alpha_T(1-\alpha_D)$ for the import share of income, so that $s_{AB} = s_{BA} = m$ at the balanced free-trade baseline with normalized incomes, and let $\nu_B \equiv d\log e_{AB}$. Suppose A imposes $d\tau = dt_{AB}$. Each country has a single import origin, so $dp_{AB} = d\log P_{M_A} = \nu_B + d\tau$ and $dp_{BA} = d\log P_{M_B} = -\nu_B$, and the cross-origin margin in (2) is inactive. Log-differentiating (2) and (3), with incomes in the common currency,

$$\begin{aligned} d\log s_{AB} &= (1-\eta)\alpha_D(\nu_B + d\tau), \qquad & d\log s_{BA} &= -(1-\eta)\alpha_D\,\nu_B, \\ d\log \widetilde{I}_A &= m\,d\tau, \qquad & d\log \widetilde{I}_B &= \nu_B. \end{aligned}$$

Substituting into (4) and using that the tariff wedge lowers the border value of A's imports by $d\tau$,

$$d\,\mathrm{TB}_A = m\Big[\big(1 + 2\alpha_D(\eta - 1)\big)\nu_B + \big(1 - m + \alpha_D(\eta - 1)\big)d\tau\Big].$$

Define

$$\rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \alpha_D\,\eta + (1 - \alpha_D)(1 - \alpha_T) \;>\; 0. \tag{8}$$

The equilibrium condition can then be written as

$$d\,\mathrm{TB}_A = \underbrace{m\,D_2\,\nu_B}_{\text{exchange-rate effect}} + \underbrace{m\,\rho_1^*\,d\tau}_{\text{tariff effect at fixed exchange rates}} = 0, \qquad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \tag{9}$$

and therefore

$$\boxed{\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\,\frac{\rho_1^*}{D_2}} \tag{10}$$

The coefficient ρ_1^* summarizes the trade-balance effect of the tariff at unchanged exchange rates. It reflects substitution between domestic and imported goods, governed by η, together with the associated income and expenditure effects. The elasticity ρ, which governs substitution across foreign import origins, is inactive with only two countries. It becomes relevant for $N \geq 3$, when a tariff on one partner can redirect import demand toward other foreign suppliers.

In reduced form, let export volume be $X(1/e)$ and import volume be $M((1+\tau)e)$, with elasticities ε_X and ε_M. Balanced trade gives $\partial\mathrm{TB}_A/\partial e = M(\varepsilon_X + \varepsilon_M - 1)$. Comparing this expression with (9) per unit of import value gives

$$\varepsilon_X + \varepsilon_M - 1 = D_2, \qquad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1.$$

Proposition 3.1 (Conventional wisdom, $N = 2$)**.** *At $N = 2$, an appreciation restores trade balance after a tariff if and only if the textbook Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$ holds. In our nested-CES model this condition is $D_2 > 0$, which holds whenever $\eta \geq 1$, and the bilateral exchange-rate response is $d\log e_{AB}/d\tau = -\rho_1^*/D_2 < 0$.*

Appendix A gives the exact finite-tariff Cobb–Douglas solution.

4 From Two to N Countries: General Exchange-Rate Adjustments

The following results require only that trade balances are differentiable with respect to exchange rates and tariffs, and do not depend on our specific nested-CES model structure.

4.1 The local disturbance–adjustment representation

Consider any differentiable multilateral trade environment with $N - 1$ independent trade-balance conditions, $\mathrm{TB}_i = 0$ for $i \neq A$, which are each functions of exchange rates and tariffs. Stack the exchange-rate changes as $\nu \equiv (\nu_j)_{j\neq A}$ and the bilateral tariff changes as $t \equiv (t_{jk})_{j\neq k}$ with $t_{jk} \equiv dt_{jk}$, so that ν and t are the perturbation vectors of the levels e_{Aj} and τ_{jk}. The conditions for $i \neq A$ are

$$J\,\nu + G\,t = 0, \qquad J_{il} \equiv \frac{\partial\, d\mathrm{TB}_i}{\partial \nu_l}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial\, d\mathrm{TB}_i}{\partial t_{jk}}\right|_{\nu \text{ fixed}}. \tag{11}$$

A tariff experiment fixes a direction a and a scalar size $d\tau$, $t = a\,d\tau$. Defining $F \equiv Ga$, the equilibrium conditions become

$$J\,\nu = -F\,d\tau, \qquad \nu = (-J)^{-1}F\,d\tau. \tag{12}$$

F gives the trade-balance effects of the tariff at unchanged exchange rates, including substitution and income effects, while J describes how trade balances respond to exchange-rate changes. At $N = 2$, $J = -mD_2$ and $F = m\rho_1^*$ in (9).

4.2 The multilateral Marshall–Lerner condition

At $N = 2$, the requirement that exchange-rate adjustment stabilizes the trade balance is the scalar condition $D_2 > 0$ of Section 3, i.e. $-J = mD_2 > 0$. The N-country generalization requires that the adjustment mapping $(-J)^{-1}$ be well defined and positive.

Definition 4.1 (Multilateral Marshall–Lerner condition)**.** *The adjustment mechanism satisfies the* multilateral Marshall–Lerner condition *(MML) if*

$$-J \textit{ is a nonsingular M-matrix:} \qquad J_{il} \geq 0 \;\; (l \neq i) \quad \textit{and} \quad (-J)^{-1} \textit{ exists with } (-J)^{-1} \geq 0. \tag{13}$$

By (12), under MML, each exchange-rate response is a nonnegative-weighted combination of the trade-balance effects at fixed exchange rates. A uniformly signed F therefore cannot generate exchange-rate responses of the opposite sign. When F has mixed signs, the response of any individual exchange rate depends on the full vector of trade-balance effects rather than on the corresponding element F_i alone. The condition also implies $\det(-J) > 0$, which is used in the threshold calculations below.[1] At $N = 2$, the system is scalar and the condition reduces to $D_2 > 0$, which Section 3 showed to be the textbook elasticity condition.

The condition above is stated directly on J, but it can alternatively be obtained from a restriction on the full bilateral Jacobian. Let $\widetilde{J}$ be the full $N \times N$ Jacobian, so that J is $\widetilde{J}$ with row and column A deleted. Two identities hold for $\widetilde{J}$. Each trade balance is homogeneous of degree one in currency values, so by Euler's theorem row i sums to TB_i, which is zero at a balanced baseline; and Walras' law, $\sum_i \mathrm{TB}_i \equiv 0$, ensures that the columns sum to zero. Thus

$$\sum_l \widetilde{J}_{il} = 0 \quad \forall i, \qquad\qquad \sum_i \widetilde{J}_{il} = 0 \quad \forall l. \tag{14}$$

[1] Given the Z-matrix sign pattern, the condition is further equivalent to $\mathrm{Re}\,\lambda(J) < 0$, i.e., to local stability of the tatonnement $\dot{\nu}_i = k_i \mathrm{TB}_i$ for any $k_i > 0$, so simple dynamic exchange-rate adjustment processes converge to the equilibrium.

The row identity pins down each diagonal entry of $\widetilde{J}$ from its off-diagonal entries, so a sign restriction on the off-diagonal entries alone determines the own effects and, after deleting A, the dominance of J. This gives a sufficient condition for MML.

Assumption 1 (Gross substitutability)**.** $\widetilde{J}_{il} \geq 0$ *for all* $l \neq i$.

An appreciation of currency l raises the price of l's goods, and Assumption 1 says that this improves every other country's trade balance: tradables are gross substitutes across countries. With the row identity in (14), $\widetilde{J}_{ii} = -\sum_{l \neq i} \widetilde{J}_{il} \leq 0$, and deleting row and column A gives

$$|J_{ii}| - \sum_{l \neq i,\, A} J_{il} = \widetilde{J}_{iA} \;\geq\; 0, \tag{15}$$

so J is diagonally dominant, strictly in every row with $\widetilde{J}_{iA} > 0$.

Proposition 4.2 (Sufficiency of gross substitutability)**.** *Under Assumption 1, with J irreducible and $\widetilde{J}_{iA} > 0$ for some i, MML* (13) *holds.*

Corollary 4.3 (Sign rule)**.** *Under* (13),

$$\operatorname{sign}\!\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\!\left(\sum_j \vartheta_{ij} F_j\right), \qquad \vartheta_{ij} \equiv \big[(-J)^{-1}\big]_{ij} \geq 0.$$

Here ϑ_{ij} measures the response of currency i to a unit trade-balance disturbance in country j. Under MML these weights are nonnegative, so the sign of each bilateral exchange-rate response is determined by the full vector of trade-balance effects at fixed exchange rates and the way those effects are transmitted through the exchange-rate system.

4.3 Symmetric partners

Restrict attention to tariffs imposed by A, $t = (t_j)_{j \neq A}$ with $t_j \equiv t_{Aj}$, so that G in (11) is $(N-1) \times (N-1)$. Suppose that A's trading partners are symmetric: relabeling them leaves the reduced system unchanged, $PJP' = J$ and $PGP' = G$ for every permutation matrix P of the $N-1$ partners. This holds when the partners have identical sizes, trade weights, and parameters at the baseline; A itself need not be symmetric with them. Take $N \geq 3$.

Under partner symmetry, the tariff vector can affect exchange rates through only two components: the average tariff across partners and differences in tariffs across partners. Decompose

$$t = \bar{t}\,\mathbf{1} + \tilde{t}, \qquad \bar{t} \equiv \frac{1}{N-1}\mathbf{1}'t, \qquad \mathbf{1}'\tilde{t} = 0. \tag{16}$$

We refer to $\bar{t}\,\mathbf{1}$ as the *average tariff component* and to $\tilde{t}$ as the *relative tariff component*. The latter records each partner's tariff relative to the average tariff imposed by A.

To see why this decomposition is useful, note that any matrix invariant to relabeling the partners has the form $aI + b\,\mathbf{1}\mathbf{1}'$. With $\mathbf{1}'\mathbf{1} = N-1$ and $\mathbf{1}'\tilde{t} = 0$,

$$(aI + b\,\mathbf{1}\mathbf{1}')\,t = \big(a + (N-1)b\big)\,\bar{t}\,\mathbf{1} + a\,\tilde{t}.$$

The matrix therefore acts separately on the average and relative tariff components. The corresponding eigenvalues are $a + (N-1)b$ and a. Since the relative tariff component sums to zero, it also has no effect on the equally weighted average of the bilateral exchange rates.

Lemma 4.4 (Symmetric partners)**.** *If the partners are symmetric, $-J$ and G take the form*

$$\begin{aligned} -J &= a_J I + b_J\,\mathbf{11'}: \quad j_S \equiv a_J + (N-1)\,b_J, \quad j_D \equiv a_J, \\ G &= a_G I + b_G\,\mathbf{11'}: \quad g_S \equiv a_G + (N-1)\,b_G, \quad g_D \equiv a_G. \end{aligned} \tag{17}$$

If $j_S, j_D \neq 0$, the response to any tariff vector is

$$\nu = \lambda_S\,\bar{t}\,\mathbf{1} + \lambda_D\,\tilde{t}, \qquad d\nu_A^E = \lambda_S\,\bar{t}, \qquad \nu_j - d\nu_A^E = \lambda_D\,\tilde{t}_j, \qquad \lambda_S \equiv \frac{g_S}{j_S}, \quad \lambda_D \equiv \frac{g_D}{j_D},$$

with effective-rate weights $w_{Aj} = 1/(N-1)$ implied by symmetry. MML (13) *holds if and only if*

$$j_S > 0 \qquad \textit{and} \qquad j_D \geq j_S. \tag{18}$$

Lemma 4.4 separates the exchange-rate response into an average component and a relative component. The average tariff $\bar{t}$ determines the common movement in A's bilateral exchange rates and hence its effective exchange rate. The relative tariff $\tilde{t}_j$ determines how the bilateral rate against partner j differs from that common movement. Section 5 computes these responses for the nested-CES model.

5 Tariff Responses at the Symmetric Benchmark

We now specialize the general results of Section 4 to the symmetric nested-CES model. This yields closed-form responses to an arbitrary vector of tariffs imposed by A. We then consider a unilateral tariff, tariffs on multiple partners, and the three-country case.

5.1 The linearized system

Linearizing at a balanced free-trade baseline with normalized incomes, define the home share $h_i = \alpha_D$, within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = m\,b_{ij}$ with $m \equiv \alpha_T(1-\alpha_D)$, and flows $f_{ij} = s_{ij}\widetilde{I}_i$, where $\sum_k f_{ki} = \sum_j f_{ij}$. All incomes and flows are measured in the common currency. Thus, $d\log \widetilde{I}_i$ includes the valuation term ν_i in addition to tariff revenue, while wages remain constant in local currency. Perturbing this baseline with tariffs dt_{ij} gives

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \qquad (dp_{ii} = 0), \tag{19}$$

$$d\log P_{M_i} = \sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{T_i} = (1-\alpha_D)\, d\log P_{M_i}, \tag{20}$$

$$d\log s_{ii} = -(1-\eta)(1-\alpha_D)\, d\log P_{M_i}, \tag{21}$$

$$d\log s_{ij} = (1-\rho)\big(dp_{ij} - d\log P_{M_i}\big) + (1-\eta)\alpha_D\, d\log P_{M_i}, \tag{22}$$

$$d\log \widetilde{I}_i = \nu_i + \sum_{j\neq i} s_{ij}\, dt_{ij}, \tag{23}$$

$$d\,\mathrm{TB}_i = \sum_{k\neq i} f_{ki}\big(d\log s_{ki} + d\log \widetilde{I}_k - dt_{ki}\big) - \sum_{j\neq i} f_{ij}\big(d\log s_{ij} + d\log \widetilde{I}_i - dt_{ij}\big). \tag{24}$$

These equations determine the model-specific matrices J and G in (11). At $N = 2$ they reduce to the calculation in Section 3.

At the symmetric baseline, every off-diagonal element of the full $N \times N$ exchange-rate Jacobian is

$$\widetilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \qquad D_N \equiv N\alpha_D(\eta - 1) + (N-2)\rho + 1. \tag{25}$$

Lemma E.1 in Appendix E gives the derivation. MML holds if and only if $D_N > 0$. In particular,

$$\eta \geq 1 \quad \Longrightarrow \quad D_N > 0 \quad \Longleftrightarrow \quad \text{MML (13) under symmetry.}$$

Thus D_N is the N-country counterpart of the two-country adjustment coefficient $D_2 = \varepsilon_X + \varepsilon_M - 1$.

5.2 Exchange-rate response to a tariff vector

Consider a vector $t = (t_j)_{j\neq A}$ of tariffs imposed by A, decomposed as in (16) into its average $\bar{t}$ and relative component $\tilde{t}$.

At unchanged exchange rates, the effect on partner i's trade balance follows from (24). Since only i's exports to A are directly affected,

$$\begin{aligned} d\,\mathrm{TB}_i\big|_{\nu \text{ fixed}} &= f_{Ai}\big(d\log s_{Ai} + d\log \widetilde{I}_A - t_i\big) \\ &= \frac{m}{N-1}\Big[-\rho_1^*\,\bar{t} - \rho\,(t_i - \bar{t})\Big]. \end{aligned} \tag{26}$$

Hence the coefficients associated with the average and relative tariff components are

$$g_S = -\frac{m\rho_1^*}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}. \tag{27}$$

Deleting row and column A from the full Jacobian gives

$$-J = c\big(NI - \mathbf{1}\mathbf{1}'\big), \qquad c \equiv \frac{mD_N}{(N-1)^2}, \qquad j_S = c, \qquad j_D = Nc. \tag{28}$$

Combining (27) and (28),

$$\lambda_S = \frac{g_S}{j_S} = -\frac{(N-1)\rho_1^*}{D_N}, \qquad \lambda_D = \frac{g_D}{j_D} = -\frac{(N-1)\rho}{ND_N}. \tag{29}$$

Theorem 5.1 (Exchange-rate response to a tariff vector)**.** *At the symmetric benchmark with $N \geq 3$ and $D_N > 0$, the response to any vector of tariffs imposed by A is*

$$\nu_j = \lambda_S \bar{t} + \lambda_D \tilde{t}_j, \qquad d\nu_A^E = \lambda_S \bar{t}, \qquad \nu_j - d\nu_A^E = \lambda_D \tilde{t}_j, \tag{30}$$

where λ_S and λ_D are given in (29)*. For any two partners $j, k \neq A$,*

$$d \log e_{jk} = \lambda_D (t_k - t_j).$$

The average tariff therefore determines the effective exchange-rate response, while tariff differentials determine bilateral deviations around it. In particular, relative tariff changes that leave $\bar{t}$ unchanged do not affect A's effective exchange rate.

5.3 Application: a unilateral tariff

Suppose A raises its tariff on B alone, so that $t = d\tau\, \mathbf{e}_B$. Then

$$\bar{t} = \frac{d\tau}{N-1}, \qquad \tilde{t}_B = \frac{N-2}{N-1} d\tau, \qquad \tilde{t}_C = -\frac{d\tau}{N-1} \quad \text{for each bystander } C.$$

Substituting into Theorem 5.1 gives

$$\frac{d \log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N} < 0, \qquad \frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{ND_N}, \tag{31}$$

and

$$\frac{d \log e_{BC}}{d\tau} = \frac{(N-1)\rho}{ND_N} > 0.$$

Corollary 5.2 (Bilateral ambiguity)**.** *For $t = d\tau\, \mathbf{e}_B$ under $D_N > 0$, A appreciates against the target B, while it depreciates against a bystander C if and only if*

$$\boxed{\rho > \rho_N^* \equiv N\rho_1^*}.$$

The effective exchange rate has an unambiguous response:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = -\frac{\rho_1^*}{D_N} < 0. \tag{32}$$

Corollary 5.3 (Effective exchange rate)**.** *For $t = d\tau\, \mathbf{e}_B$ under $D_N > 0$, A's effective exchange rate appreciates for every $N \geq 2$ and every admissible ρ.*

The bilateral reversal threshold differs from the condition under which the bystander initially gains trade. At unchanged exchange rates, (26) gives

$$F_B = -\frac{m\big[\rho_1^* + (N-2)\rho\big]}{(N-1)^2} < 0, \qquad F_C = \frac{m(\rho - \rho_1^*)}{(N-1)^2}. \tag{33}$$

Thus the tariff improves C's trade balance at fixed exchange rates whenever

$$\rho > \rho_1^*,$$

whereas the bilateral exchange rate reverses only when

$$\rho > N\rho_1^*.$$

The difference arises from equilibrium exchange-rate adjustment. Under symmetry, (28) implies $j_D/j_S = N$: the relative component of a trade-balance disturbance induces stronger adjustment than the common component. The factor N in the bilateral threshold therefore comes from the exchange-rate system, not from the initial trade-balance effect of the tariff.[2]

A useful restriction follows when the two substitution elasticities coincide.

Corollary 5.4 (Single elasticity)**.** *With $\rho = \eta = \sigma$,*

$$\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \geq 0$$

whenever $\alpha_D \geq 1/N$, with equality only at $\alpha_D = 1/N$ and $\alpha_T = 1$. Hence the bystander reversal is impossible in this region. Without imposing $\rho = \eta$, the same restriction implies

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \geq 0,$$

so $\rho > \eta$ is necessary, but not sufficient, for a bystander reversal. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$,

$$\frac{d\log e_{AC}}{d\tau} = -\frac{1-\alpha_T}{N\sigma},$$

which approaches zero from below as $\alpha_T \to 1$.

Appendix B gives the corresponding real exchange-rate results.

5.4 Application: multiple targets

Theorem 5.1 also orders bilateral responses directly by tariff treatment.

Corollary 5.5 (Relative tariff treatment)**.** *Under the conditions of Theorem 5.1, with $\bar{t} > 0$, $t_j > t_k$ implies $\nu_j < \nu_k$. Moreover, A depreciates against partner j if and only if*

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t} \qquad \Longleftrightarrow \qquad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}.$$

[2] The exact factor N is specific to the symmetric nested-CES benchmark. Appendix C considers asymmetric baselines.

Thus A appreciates more against partners facing higher tariffs. Depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_1^*$ and a sufficiently large tariff differential.

If A raises its tariff by τ on a set of k partners, the same decomposition gives the following result.

Corollary 5.6 (Number of targets)**.** *If A raises its tariff by τ on k partners, $1 \leq k \leq N-1$, then for every untargeted partner C and every targeted partner B,*

$$\frac{d\log e_{AC}}{\tau} = \frac{k(\rho - N\rho_1^*)}{ND_N}, \qquad \frac{d\log e_{AB}}{\tau} = -\frac{k\rho_1^*}{D_N} - \frac{(N-1-k)\rho}{ND_N}, \qquad \frac{d\nu_A^E}{\tau} = -\frac{k\rho_1^*}{D_N}. \tag{34}$$

For every $k \leq N-2$, the sign of the bystander response is therefore determined by the same threshold $\rho > N\rho_1^*$.

When all partners face the same tariff,

$$\frac{d\log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \qquad \forall j. \tag{35}$$

All bilateral rates then move together, and substitution across foreign origins plays no role.

5.5 The three-country case

For $N = 3$, the unilateral-tariff system can be written explicitly as

$$J = -\frac{mD_3}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \qquad D_3 = 3\alpha_D(\eta - 1) + \rho + 1. \tag{36}$$

Hence

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1}F\,d\tau = \frac{1}{3D_3}\begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau.$$

Proposition 5.7 (Three-country threshold)**.** *With $D_3 > 0$,*

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \qquad \boxed{\frac{d\log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}}, \qquad \frac{d\log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0. \tag{37}$$

Thus A appreciates against B, B depreciates against C, and A depreciates against C if and only if $\rho > 3\rho_1^*$.

The three-country case also makes the distinction between the initial trade-balance effect and the equilibrium exchange-rate response transparent. The bystander's trade balance improves at fixed exchange rates when $\rho > \rho_1^*$, while its bilateral exchange rate reverses only when $\rho > 3\rho_1^*$. By Corollary 4.3,

$$\frac{d\nu_C}{d\tau} \propto F_B + 2F_C \propto \rho - 3\rho_1^*.$$

At $N = 3$, Corollary 5.4 implies that the reversal is impossible under a common elasticity $\rho = \eta$ whenever $\alpha_D \geq 1/3$.

6 Trade Wars and Preferential Tariffs

The results above also apply when several countries change tariffs simultaneously. Under symmetry, the bilateral exchange-rate effects can be computed for each tariffing country using Theorem 5.1 and then added.

6.1 Trade wars

Bilateral retaliation. Suppose A raises its tariff on B by $d\tau$ and B raises its tariff on A by the same amount. The two effects on the A–B rate cancel, while for any bystander C,

$$\left.\frac{d\log e_{AB}}{d\tau}\right|^{w} = 0, \qquad \left.\frac{d\log e_{AC}}{d\tau}\right|^{w} = \frac{\rho - \rho_1^*}{D_N}, \qquad \left.\frac{d\nu_A^E}{d\tau}\right|^{w} = \frac{N-2}{N-1}\,\frac{\rho - \rho_1^*}{D_N}. \tag{38}$$

Proposition 6.1 (Trade-war threshold)**.** *Under MML, in a symmetric bilateral trade war the belligerents' effective exchange rates depreciate, and every bystander appreciates against both, if and only if*

$$\rho > \rho_1^*.$$

The threshold is independent of N.

Relative to the unilateral tariff, retaliation lowers the bystander threshold from $N\rho_1^*$ to ρ_1^*. The reason is that B's retaliation introduces an additional relative tariff effect on the A–C rate without adding an average tariff effect from B to that rate. Using (29),

$$\left.\frac{d\log e_{AC}}{d\tau}\right|^{w} = -\frac{\rho_1^*}{D_N} + \frac{\rho}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}. \tag{39}$$

Thus the factor N that appears in the unilateral-tariff threshold is absent under symmetric retaliation.

Global retaliation. Suppose instead that A raises its tariff by $d\tau$ on every partner and each partner raises its tariff by $d\tau$ on A.

Proposition 6.2 (Global retaliation)**.** *Under MML,*

$$\frac{d\log e_{Aj}}{d\tau} = \frac{d\nu_A^E}{d\tau} = \frac{(N-2)(\rho - \rho_1^*)}{D_N} \qquad \forall j,$$

while all exchange rates among A's partners are unchanged. Hence A depreciates against every partner if and only if $\rho > \rho_1^*$.

The response vanishes at $N = 2$ by symmetry. For $N \geq 3$, the same threshold as in the bilateral trade war applies.

6.2 Competing exporters

A standard benchmark in the preferential-trade-agreement literature has one importing country and two competing exporters, following Viner (1950) and Mundell (1964). Importer A is endowed only with the numeraire good y, while exporters B and C are endowed with good x. Preferences are quasilinear, $U_i = u_i(x_i) + y_i$. Country A levies specific tariffs t_B and t_C. Let q_j denote exporter j's price and p the domestic price in A. Market clearing requires

$$p = q_B + t_B = q_C + t_C, \qquad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \qquad M'_A < 0, \quad E'_j > 0, \tag{40}$$

where $M_A(p) = x^d_A(p)$ and $E_j(q_j) = \omega_j - x^d_j(q_j)$. Differentiating with respect to t_B gives

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0,1), \qquad \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \qquad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \tag{41}$$

Proposition 6.3 (Competing-exporter terms-of-trade effect)**.** *A tariff on imports from B worsens B's terms of trade and improves C's terms of trade.*

6.3 Preferential tariffs

The competing-exporter result provides a useful comparison with the three-country case of our model. An improvement in C's terms of trade corresponds to an increase in e_{AC} by (7). In the competing-exporter model this occurs unconditionally after a tariff on B, whereas in our model Proposition 5.7 requires

$$\rho > 3\rho^*_1.$$

The difference reflects the economic environment. In the competing-exporter model, A does not produce the importable good, so a tariff reallocates demand between the two foreign suppliers without inducing substitution toward a domestically produced import-competing good. In our model, both margins are present: substitution between domestic and imported goods, governed by η, and substitution across foreign import origins, governed by ρ. The terms-of-trade effect on a third country therefore depends on their relative strength.

A preferential tariff reduction gives the corresponding result with the signs reversed. Suppose A lowers its tariff on B alone, so that $t = -\tau\, \mathbf{e}_B$. For a non-member C, Theorem 5.1 implies that C's terms of trade worsen if and only if

$$\rho > N\rho^*_1.$$

Thus the standard competing-exporter result emerges only when substitution across foreign origins is sufficiently strong relative to the other general-equilibrium effects in our model.

The two environments are not formally nested. The competing-exporter model has a common importable supplied by B and C and quasilinear preferences, while our model has country-specific differentiated tradables and nested-CES preferences. The comparison is therefore one of economic mechanisms rather than parameter restrictions. In particular, the competing-exporter model isolates substitution across foreign suppliers, whereas our framework allows this effect to interact with domestic-import substitution and income effects.

At $N = 3$, the threshold can be written as

$$\rho_3^* = 3\rho_1^* = 3 + 3\alpha_D(\eta - 1) - 3\alpha_T(1 - \alpha_D). \tag{42}$$

The threshold rises with the strength of substitution toward the domestic good and falls with the import share. This decomposition makes clear why the terms-of-trade effect that is unambiguous in the competing-exporter benchmark can become ambiguous in the nested-CES environment.

Competing-exporter models such as Bagwell and Staiger (1999) and Ornelas (2005) abstract from trade between the foreign suppliers and use quasilinear preferences, while bloc and endowment models such as Krugman (1991) and Bond and Syropoulos (1996) abstract from domestic production of the importable. Kemp and Wan (1976) consider a more general customs-union environment. The comparison here is not intended to nest these models, but to identify which additional margins determine the sign of third-country terms-of-trade effects in our setting.

7 Conclusion

A tariff affects exchange rates through two distinct steps. At unchanged exchange rates it changes countries' trade balances, summarized by F; exchange rates then adjust through the system $(-J)^{-1}$. In the two-country case this reduces to the familiar Marshall–Lerner result: under the elasticity condition, a tariff appreciates the tariffing country's currency.

With three or more countries, a tariff can also redirect expenditure across foreign suppliers. As a result, the tariffing country continues to appreciate against the targeted partner, but its exchange rate against an untariffed partner can move in either direction. Under symmetry, the response to any tariff vector separates into an average tariff component and a relative tariff component. The average tariff determines the effective exchange-rate response, while tariff differentials determine the dispersion of bilateral responses around it.

For a unilateral tariff on one partner, the tariffing country's effective exchange rate always appreciates under MML. Its bilateral rate against an untariffed bystander reverses only if

$$\rho > N\rho_1^*.$$

The corresponding trade-balance effect on the bystander already changes sign at $\rho > \rho_1^*$. The difference between these thresholds arises from multilateral exchange-rate adjustment: under the symmetric nested-CES benchmark, the relative component of a trade-balance disturbance induces an adjustment N times as strong as the common component.

Other tariff configurations change this comparison. Under symmetric bilateral retaliation, the bystander threshold falls to ρ_1^* and is independent of N. Preferential tariffs yield the same distinction from another direction. In the standard competing-exporter benchmark, substitution across foreign suppliers generates an unambiguous third-country terms-of-trade effect. In our model that effect competes with substitution between domestic and imported goods and with income effects, producing a threshold rather than an unconditional sign.

The main implication is that the familiar two-country exchange-rate result survives multilateralization most cleanly at the level of the effective exchange rate. Bilateral exchange rates contain additional information about how trade policy differs across partners, and their responses need not inherit the sign of the aggregate adjustment.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$, expenditure shares are fixed at their weights. From (3), $I_A = w_A L_A/[1 - \frac{\tau}{1+\tau} m]$, and $\mathrm{TB}_A = 0$ becomes $e\, m\, w_B L_B = m I_A/(1+\tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1-m)\tau}, \qquad \left.\frac{d \log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_1^*}{D_2}\right|_{\eta=1}. \tag{43}$$

Thus, $e(\tau)$ is decreasing in τ, so the sign of the linearized response in (10) is exact for all tariff levels in this case. Here $\eta = 1$ fixes the elasticity, not the weight α_D.

B Real exchange rates

This appendix gives the real exchange-rate counterparts of the results in Sections 5.2–5.3.

From (5) and (1),

$$d \log q_{Aj} = \nu_j + \alpha_T \big(d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}} \big).$$

For a tariff vector t, symmetry implies

$$d \log P_{M_A} = d\nu_A^E + \bar{t}, \qquad d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} \nu_j$$

for a partner that imposes no tariff. Hence

$$\begin{aligned} d\log q_{Aj} &= \Big(1-\frac{mN}{N-1}\Big)\nu_j - m\,\bar{t}, \\ dq_A^E &= \Big[\Big(1-\frac{mN}{N-1}\Big)\lambda_S - m\Big]\bar{t}, \qquad (44) \\ d\log q_{Aj} - dq_A^E &= \Big(1-\frac{mN}{N-1}\Big)\lambda_D\,\tilde{t}_j. \end{aligned}$$

Thus the average tariff component determines the real effective exchange-rate response, while the relative tariff component determines bilateral deviations around it, subject to the price-index adjustment in (44).

For a unilateral tariff, $\bar{t} = d\tau/(N-1)$ and

$$d\log q_{Aj} = \Big(1-\frac{mN}{N-1}\Big)\nu_j - \frac{m}{N-1}\,d\tau, \qquad j = B, C. \tag{45}$$

Setting $d\log q_{AC} = 0$ and using (31), for $m < 1/N$,

$$\rho_q^*(N) = \frac{\big[(N-1) - mN\big]\,N\rho_1^* + mN\big[N\alpha_D(\eta-1)+1\big]}{(N-1)(1-mN)} \;>\; N\rho_1^*. \tag{46}$$

Averaging (45) gives

$$\frac{dq_A^E}{d\tau} = -\Big(1-\frac{mN}{N-1}\Big)\frac{\rho_1^*}{D_N} - \frac{m}{N-1} \;<\; 0 \qquad \text{for all } \rho, N\ \big(m < \tfrac{N-1}{N}\big). \tag{47}$$

Proposition B.1 (Real exchange rates)**.** *Under MML, (i) the real bilateral reversal threshold exceeds the nominal threshold, $\rho_q^*(N) > N\rho_1^*$, and (ii) the REER appreciates for every ρ and N, $dq_A^E/d\tau < 0$.*

Both results reflect the direct price-index effect of the tariff. At a given nominal exchange rate, the tariff raises A's consumer prices, making a real bilateral depreciation harder to obtain and reinforcing the effective appreciation.

C Asymmetry

The disturbance-adjustment representation (12) and MML (13) do not require symmetry. The closed-form thresholds, the NEER cancellation, and the average and relative tariff decomposition in Sections 5–6 do. This appendix examines which results survive when partner symmetry is relaxed.

A simple degree argument shows why the single-threshold results need not extend to arbitrary trade networks. The elasticity ρ enters the share responses (22) only through $(1-\rho)$, so every entry of J and F is linear in ρ. Each entry of $\mathrm{adj}(-J)$ is a determinant of an $(N-2)\times(N-2)$ submatrix of $-J$ and is therefore a polynomial of degree at most $N-2$ in ρ. It follows that each numerator $[\mathrm{adj}(-J)F]_i$ in (12) is a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under

MML, a bilateral response can change sign only at a root of this numerator polynomial. Symmetry reduces the problem to the two invariant components of Section 4.3, yielding the linear thresholds in Proposition 5.7 and Corollary 5.2. At a general asymmetric baseline, the $N = 3$ numerator is already quadratic in ρ, as shown below, and for $N \geq 4$ a response may change sign more than once as ρ varies.

General asymmetric baseline. At any balanced baseline, (19)–(24) continue to hold with trade shares (b_{ij}, h_i) and incomes y_i as coefficients. For $N = 3$, the bystander slope has denominator $\det(-J) > 0$ and a numerator that is quadratic in ρ:

$$\operatorname{sign} \frac{d \log e_{AC}}{d\tau} = \operatorname{sign}\left(c_2\rho^2 + c_1\rho + c_0\right), \qquad c_2 = b_{AB}b_{AC}b_{CA}b_{CB}(1-h_A)(1-h_C)y_A y_C. \tag{48}$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. The large-ρ sign is economically relevant only if MML continues to hold over the corresponding parameter range, ensuring $\det(-J) > 0$. Under this condition, $c_2 > 0$ whenever A and C each import from both foreign origins. Hence sufficiently strong substitution across import origins eventually reverses e_{AC} at any such baseline. The relevant threshold is the largest positive root and equals $3\rho_1^*$ at the symmetric point. Away from symmetry, however, there is no universal bilateral threshold.

Structured asymmetry. To examine departures from symmetry while preserving tractability, let the target receive a multiple κ of its symmetric bilateral import share. This parameterization remains comparable across N because the symmetric share falls as $1/(N-1)$. Assume the weight matrix is symmetric and balanced:

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \qquad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \qquad b_{CC'} = \frac{1 - 2b_{CA}}{N-3}. \tag{49}$$

The bystanders remain interchangeable, so the reduced system again has two unknowns and can be solved in closed form. With NEER weights $w_{Aj} = b_{Aj}$, the response is a rational function of (ρ, N, κ) that reduces to $-\rho_1^*/D_N$ at $\kappa = 1$.

Proposition C.1 (Asymmetric NEER)**.** *Holding κ fixed as $N \to \infty$,*

$$N \frac{d\nu_A^E}{d\tau} \longrightarrow -\kappa \frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} = -\kappa \lim_{N\to\infty} \frac{N\rho_1^*}{D_N}. \tag{50}$$

To leading order, asymmetric exposure scales the symmetric NEER response by the relative importance of the target without changing its sign. At finite N, the exact numerator can be written as $-\kappa\rho_1^*$ times a quadratic $Q(\rho)$. If $\kappa \geq 1$, all coefficients of Q are positive for sufficiently large N, so the effective rate appreciates for every ρ. If $\kappa < 1$, Q has a positive root satisfying

$$\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1-\kappa}\left(1 + O(N^{-1})\right),$$

above which the NEER depreciates. When the target receives less than its symmetric trade weight, depreciations against the bystanders can therefore outweigh the appreciation against the target. As $\kappa \to 0$, $\rho_E^* \to N\rho_1^*$, so the reversal threshold remains of order N.

Under asymmetry, G and $-J$ generally no longer share the constant and zero-sum eigenspaces of Lemma 4.4. Relative tariff changes can therefore affect the effective exchange rate. The NEER result is exact under symmetry and approximately robust in the structured asymmetric family above; no corresponding claim is made for arbitrary asymmetric trade networks.

D Correspondence across N

object	$N = 2$	$N = 3$	general N
adjustment condition	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	MML: $-J$ a nonsingular M-matrix
sufficient primitive condition	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 1)
sign rule	scalar response	weights $\frac{1}{3}\left(\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right)$	$\operatorname{sign}\left(\sum_j \vartheta_{ij} F_j\right)$, $\vartheta \geq 0$
adjustment eigenvalues j_S, j_D (28)	mD_2, —	$mD_3/4$, $3mD_3/4$	$mD_N/(N-1)^2$, $NmD_N/(N-1)^2$
tariff-effect eigenvalues g_S, g_D (27)	$-m\rho_1^*$, —	$-m\rho_1^*/2$, $-m\rho/2$	$-m\rho_1^*/(N-1)$, $-m\rho/(N-1)$
bilateral reversal threshold	none	$3\rho_1^*$	$N\rho_1^*$ under symmetry; polynomial roots in general
real bilateral threshold	none	$\rho_q^*(3)$	$\rho_q^*(N)$ (46)
NEER, unilateral tariff	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$ under symmetry
response coefficients λ_S, λ_D (29)	$-\rho_1^*/D_2$, —	$-2\rho_1^*/D_3$, $-2\rho/(3D_3)$	$-(N-1)\rho_1^*/D_N$, $-(N-1)\rho/(ND_N)$
trade-war threshold	—	ρ_1^*	ρ_1^*, independent of N

E Proofs

Throughout, $m \equiv \alpha_T(1-\alpha_D)$, $P \equiv \rho + \alpha_D(\eta - 1)$, and all derivatives are evaluated at the balanced free-trade baseline of Section 5.1.

Proof of Proposition 3.1. The linearization in Section 3 gives (9), with $D_2 = 1 + 2\alpha_D(\eta - 1)$ and $\rho_1^* = 1 - m + \alpha_D(\eta - 1)$, and (10) follows. For the elasticity form, hold income fixed and let $M(p)$

denote import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB} I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 - (1-\eta)\alpha_D = 1 + \alpha_D(\eta - 1).$$

By symmetry, $\varepsilon_X = 1 + \alpha_D(\eta - 1)$. Differentiating $\mathrm{TB}_A(e, \tau) = X(1/e) - e\, M((1+\tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives

$$\frac{\partial \mathrm{TB}_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1).$$

Hence $\varepsilon_X + \varepsilon_M - 1 = 2\alpha_D(\eta - 1) + 1 = D_2$, so the stability condition $D_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$. Since $F = m\rho_1^* > 0$ by (8), the slope $-\rho_1^*/D_2$ is negative. □

Proof of Proposition 4.2. Scaling every currency value e_{Al} by a common factor leaves all relative prices in (19) unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i, by the same factor. Thus TB_i is homogeneous of degree one in the currency values. By Euler's theorem, $\sum_l \widetilde{J}_{il} = \mathrm{TB}_i$, which is zero at the balanced baseline, so $\sum_l \widetilde{J}_{il} = 0$ for all i. Equivalently, in (24), a common shift in all ν_l affects only the income terms, whose export and import contributions cancel by baseline balance. Summing (4) over i gives $\sum_i \mathrm{TB}_i \equiv 0$, so $\sum_i \widetilde{J}_{il} = 0$ for all l. Under Assumption 1, $\widetilde{J}_{il} \geq 0$ for $l \neq i$, so $\widetilde{J}_{ii} = -\sum_{l \neq i} \widetilde{J}_{il} \leq 0$. Deleting row and column A then gives (15), with strict inequality in every row for which $\widetilde{J}_{iA} > 0$.

Thus $-J$ is a Z-matrix with nonpositive off-diagonal entries, nonnegative diagonal entries, weak diagonal dominance, irreducibility, and strict diagonal dominance in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$, where $D = \mathrm{diag}(-J) > 0$ and $B \geq 0$. Diagonal dominance implies that every row sum of $D^{-1}B$ is at most one and that at least one is strictly below one. With irreducibility, its spectral radius therefore satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1} D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \;\geq\; 0.$$

Nonsingularity follows from $r < 1$. Moreover,

$$\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0.$$

Finally, the Gershgorin discs of $-J$ are centered at $|J_{ii}| > 0$ with radii no larger than $|J_{ii}|$. They therefore lie in the closed right half-plane and can intersect the imaginary axis only at the origin, which nonsingularity excludes. Hence $\mathrm{Re}\,\lambda(-J) > 0$, or equivalently $\mathrm{Re}\,\lambda(J) < 0$. □

Proof of Corollary 4.3. The result follows directly from (12):

$$\frac{d\nu}{d\tau} = (-J)^{-1} F,$$

with $(-J)^{-1} = (\vartheta_{ij}) \geq 0$ elementwise under MML. □

Proof of Lemma 4.4. Let M be an $(N-1)\times(N-1)$ matrix with $PMP' = M$ for every permutation matrix P. Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i,j) to (k,l), $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence

$$M = aI + b\mathbf{1}\mathbf{1}'.$$

Since $\mathbf{1}'\mathbf{1} = N-1$,

$$M\mathbf{1} = [a + (N-1)b]\mathbf{1},$$

while $Mv = av$ for every v satisfying $\mathbf{1}'v = 0$. This gives (17) for $-J$ and G.

For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D}I + \frac{1}{N-1}\Big(\frac{1}{j_S} - \frac{1}{j_D}\Big)\mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the constant and zero-sum eigenspaces, with response coefficients g_S/j_S and g_D/j_D. Writing

$$t = \bar{t}\,\mathbf{1} + \tilde{t}, \qquad \mathbf{1}'\tilde{t} = 0,$$

gives

$$\nu = \frac{g_S}{j_S}\bar{t}\,\mathbf{1} + \frac{g_D}{j_D}\tilde{t},$$

and averaging with equal weights yields

$$d\nu_A^E = \frac{g_S}{j_S}\bar{t}.$$

For (18), the off-diagonal entry of $-J$ is

$$b_J = \frac{j_S - j_D}{N-1},$$

so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, the entries of $(-J)^{-1}$ above are nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or its inverse cannot be elementwise nonnegative. At $N = 2$ the zero-sum subspace is trivial and the condition reduces to $j_S > 0$. □

Lemma E.1 (Full Jacobian at the symmetric baseline)**.** *At the symmetric balanced baseline of Section 5.1, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full $N \times N$ Jacobian of the nested-CES model is*

$$\widetilde{J}_{il} = \frac{m}{(N-1)^2}D_N \quad (l \neq i), \qquad \widetilde{J}_{ii} = -\frac{m}{N-1}D_N, \qquad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1.$$

Proof. Set $dt_{jk} = 0$, so $dp_{ij} = \nu_j - \nu_i$ and $d\log\widetilde{I}_i = \nu_i$. Writing $S \equiv \sum_j \nu_j$, (20) gives

$$d\log P_{M_k} = \frac{S - N\nu_k}{N-1}, \qquad \frac{\partial\, d\log P_{M_k}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side,

$$\sum_{j\neq i}\big(dp_{ij} - d\log P_{M_i}\big) = 0,$$

so the $(1-\rho)$ terms cancel when (22) is summed over $j \neq i$. Thus,

$$\sum_{j\neq i} d\log s_{ij} = (N-1)(1-\eta)\alpha_D\, d\log P_{M_i}.$$

Substituting into (24) gives

$$d\,\mathrm{TB}_i = \frac{m}{N-1}\Big[\sum_{k\neq i}\big(d\log s_{ki} + \nu_k\big) - (N-1)(1-\eta)\alpha_D\, d\log P_{M_i} - (N-1)\nu_i\Big].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l}\sum_{k\neq i} d\log s_{ki} = -(1-\rho)\frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms gives

$$\widetilde{J}_{il} = \frac{m}{(N-1)^2}\Big[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1)\Big] = \frac{m}{(N-1)^2}D_N.$$

The diagonal follows from (14):

$$\widetilde{J}_{ii} = -(N-1)\widetilde{J}_{il} = -\frac{m}{N-1}D_N.$$

At $N = 2$ this gives $\widetilde{J}_{AB} = mD_2$, the coefficient in (9); deleting row and column A at $N = 3$ reproduces J in (36). □

Proof of Proposition 5.7. At $N = 3$, (19)–(20) give

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C - \nu_B, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B - \nu_C.$$

Also,

$$d\log \widetilde{I}_A = \tfrac{m}{2}d\tau, \qquad d\log \widetilde{I}_B = \nu_B, \qquad d\log \widetilde{I}_C = \nu_C.$$

The deviations of bilateral prices from the corresponding import indices are

$$\begin{aligned}
dp_{AB} - d\log P_{M_A} &= \tfrac{1}{2}(\nu_B - \nu_C + d\tau), & dp_{AC} - d\log P_{M_A} &= \tfrac{1}{2}(\nu_C - \nu_B - d\tau),\\
dp_{BA} - d\log P_{M_B} &= -\tfrac{1}{2}\nu_C, & dp_{BC} - d\log P_{M_B} &= \tfrac{1}{2}\nu_C,\\
dp_{CA} - d\log P_{M_C} &= -\tfrac{1}{2}\nu_B, & dp_{CB} - d\log P_{M_C} &= \tfrac{1}{2}\nu_B.
\end{aligned}$$

The share changes then follow from (22). With flow values $m/2$, (24) gives

$$\begin{aligned}
\tfrac{2}{m}\, d\,\mathrm{TB}_B &= \big(d\log s_{AB} + \tfrac{m}{2} d\tau - d\tau\big) + \big(d\log s_{CB} + \nu_C\big) - \big(d\log s_{BA} + \nu_B\big) - \big(d\log s_{BC} + \nu_B\big),\\
\tfrac{2}{m}\, d\,\mathrm{TB}_C &= \big(d\log s_{AC} + \tfrac{m}{2} d\tau\big) + \big(d\log s_{BC} + \nu_B\big) - \big(d\log s_{CA} + \nu_C\big) - \big(d\log s_{CB} + \nu_C\big).
\end{aligned}$$

For example, the coefficient on ν_B in $\frac{2}{m} d\,\mathrm{TB}_B$ is

$$\tfrac{1}{2}\big[(1-\rho) + (1-\eta)\alpha_D\big] + \tfrac{1}{2}\big[(1-\rho) + (1-\eta)\alpha_D\big] + 2(1-\eta)\alpha_D - 2 = -D_3.$$

Collecting the remaining coefficients gives (36), and inverting gives (37). With $D_3 > 0$, the numerator for e_{AB} is negative and the numerator for e_{BC} is positive. The bystander response satisfies

$$\operatorname{sign} \frac{d\log e_{AC}}{d\tau} = \operatorname{sign}(\rho - 3\rho_1^*).$$

□

Proof of Corollary 5.4. With $\rho = \eta = \sigma$,

$$\rho_N^* = N\big[1 + \alpha_D(\sigma - 1) - \alpha_T(1-\alpha_D)\big],$$

so

$$\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T).$$

For $\alpha_D \geq 1/N$ both terms are nonnegative, with both zero only at $\alpha_D = 1/N$, $\alpha_T = 1$; reversal requires $\sigma > \rho_N^*$ and therefore fails. Since ρ_1^* does not involve ρ,

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T).$$

At $\alpha_D = 1/N$,

$$\rho_N^* - \sigma = (N-1)(1-\alpha_T), \qquad D_N = (N-1)\sigma,$$

so by (31),

$$\frac{d\log e_{AC}}{d\tau} = -\frac{1-\alpha_T}{N\sigma}.$$

□

Direct derivation of (31). The unilateral slopes can also be obtained by inverting a reduced two-equation system. Since the equilibrium is unique under MML and the system is invariant to permutations of the $N-2$ bystanders, all bystanders have the same response ν_C. The equilibrium therefore reduces to the conditions for B and one representative bystander.

Equations (19)–(20) give

$$d\log P_{M_A} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \qquad d\log P_{M_B} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \qquad d\log P_{M_C} = \frac{\nu_B - 2\nu_C}{N-1}. \tag{51}$$

Also,

$$d\log \widetilde{I}_A = \frac{m}{N-1}d\tau, \qquad d\log \widetilde{I}_B = \nu_B, \qquad d\log \widetilde{I}_C = \nu_C.$$

The deviations of bilateral prices from the corresponding import indices are

$$\begin{aligned}
&dp_{AB} - d\log P_{M_A} = \tfrac{N-2}{N-1}(\nu_B + d\tau - \nu_C), && dp_{AC} - d\log P_{M_A} = \tfrac{1}{N-1}(\nu_C - \nu_B - d\tau),\\
&dp_{BA} - d\log P_{M_B} = -\tfrac{N-2}{N-1}\nu_C, && dp_{BC} - d\log P_{M_B} = \tfrac{1}{N-1}\nu_C,\\
&dp_{CA} - d\log P_{M_C} = -\tfrac{1}{N-1}\big[(N-3)\nu_C + \nu_B\big], && dp_{CC'} - d\log P_{M_C} = \tfrac{1}{N-1}(2\nu_C - \nu_B),\\
&dp_{CB} - d\log P_{M_C} = \tfrac{1}{N-1}\big[(N-2)\nu_B - (N-3)\nu_C\big].
\end{aligned}$$

Using (22) and flow values $m/(N-1)$, collecting the two trade-balance conditions gives

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \qquad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1. \tag{52}$$

The matrix has determinant N, so inversion gives (31). At $N = 3$ this reduces to (37), and at $N = 2$ to (10).

Proof of Corollary 5.2(ii). By Lemma 4.4 and (29),

$$\nu = \lambda_S \bar{t}\,\mathbf{1} + \lambda_D \tilde{t}.$$

For $t = d\tau\,\mathbf{e}_B$,

$$\bar{t} = \frac{d\tau}{N-1}, \qquad \tilde{t}_C = -\frac{d\tau}{N-1},$$

so

$$\nu_C = \frac{\lambda_S - \lambda_D}{N-1}d\tau = \frac{\rho - N\rho_1^*}{ND_N}d\tau.$$

Under MML, $D_N > 0$, so

$$\operatorname{sign}\frac{d\log e_{AC}}{d\tau} = \operatorname{sign}(\rho - N\rho_1^*).$$

The same substitution with

$$\tilde{t}_B = \frac{N-2}{N-1}d\tau$$

gives the remaining expressions in (31). □

Proof of Corollary 5.3. By Lemma 4.4,

$$d\nu_A^E = \lambda_S \bar{t}$$

with symmetric weights $w_{Aj} = 1/(N-1)$. For $t = d\tau\,\mathbf{e}_B$,

$$\bar{t} = \frac{d\tau}{N-1},$$

so

$$\frac{d\nu_A^E}{d\tau} = \frac{\lambda_S}{N-1} = -\frac{\rho_1^*}{D_N}.$$

By (8), $\rho_1^* > 0$, while $D_N > 0$ under MML. The response is therefore negative for every $N \geq 2$ and every admissible ρ. $\square$

Proof of Theorem 5.1. Under partner symmetry, the response matrix $R = (-J)^{-1}G$ has the same constant and zero-sum eigenspaces as J and G. By (27)–(29), its eigenvalues on these two subspaces are λ_S and λ_D. Therefore, for

$$t = \bar{t}\,\mathbf{1} + \tilde{t}, \qquad \mathbf{1}'\tilde{t} = 0,$$

we have

$$\nu = Rt = \lambda_S \bar{t}\,\mathbf{1} + \lambda_D \tilde{t}.$$

Averaging with weights $1/(N-1)$ gives

$$d\nu_A^E = \lambda_S \bar{t},$$

and subtracting gives

$$\nu_j - d\nu_A^E = \lambda_D \tilde{t}_j.$$

Finally,

$$d \log e_{jk} = \nu_k - \nu_j = \lambda_D(\tilde{t}_k - \tilde{t}_j) = \lambda_D(t_k - t_j).$$

Under MML, $D_N > 0$, so $\lambda_S, \lambda_D < 0$. $\square$

Proof of Corollary 5.5. From (30),

$$\nu_j - \nu_k = \lambda_D(t_j - t_k),$$

and $\lambda_D < 0$, proving the ordering result. Moreover,

$$\nu_j > 0$$

if and only if

$$\lambda_S \bar{t} + \lambda_D(t_j - \bar{t}) > 0.$$

Substituting (29) and dividing by $(N-1)/D_N > 0$ gives

$$-\rho_1^* \bar{t} - \frac{\rho}{N}(t_j - \bar{t}) > 0,$$

or

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t}.$$

For $\bar{t} > 0$, this is equivalent to

$$\frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}.$$

The right-hand side is positive only if $\rho > N\rho_1^*$. $\square$

Proof of Corollary 5.6. With $t_j = \tau$ on the k targets and zero otherwise,

$$\bar{t} = \frac{k\tau}{N-1}, \qquad \tilde{t}_B = \frac{N-1-k}{N-1}\tau$$

for a target, and

$$\tilde{t}_C = -\frac{k\tau}{N-1}$$

for a bystander. Substituting into (30) gives

$$\nu_C = \frac{k\tau}{N-1}(\lambda_S - \lambda_D) = \frac{k\tau}{ND_N}(\rho - N\rho_1^*),$$

together with the target and NEER responses in (34). For every $k \leq N-2$, the sign of the bystander response is therefore determined by $\rho - N\rho_1^*$. At $k = N-1$ there is no bystander and $\tilde{t} = 0$, giving (35). □

Proof of Proposition 6.1. Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (31) gives

$$d\log e'_{BA} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N}, \qquad d\log e'_{BC} = \frac{\rho - N\rho_1^*}{ND_N}.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d\log e'_{AB} = \frac{N\rho_1^* + (N-2)\rho}{ND_N}, \qquad d\log e'_{AC} = \frac{(N-1)\rho}{ND_N}.$$

By linearity,

$$d\log e_{AB}\big|^{w} = 0,$$

and

$$d\log e_{AC}\big|^{w} = \frac{\rho - N\rho_1^*}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

By symmetry between A and B, the same expression applies to $d\log e_{BC}|^{w}$. Averaging gives

$$\frac{d\nu_A^E}{d\tau}\bigg|^{w} = \frac{N-2}{N-1}\frac{\rho - \rho_1^*}{D_N}.$$

Under MML, $D_N > 0$, so the signs are determined by $\rho - \rho_1^*$. □

Proof of Proposition 6.2. Superpose the N tariff vectors. A's tariff vector is uniform,

$$\bar{t}^{(A)} = d\tau, \qquad \tilde{t}^{(A)} = 0,$$

and contributes $\lambda_S d\tau$ to every $d\log e_{Aj}$.

Partner j taxes only A. Its contribution to $d\log e_{Aj}$ is

$$-\big[\lambda_S + (N-2)\lambda_D\big]\frac{d\tau}{N-1},$$

while each of the $N-2$ other partners contributes $-\lambda_D d\tau$. Summing,

$$\begin{aligned} d\log e_{Aj} &= \lambda_S d\tau - \big[\lambda_S + (N-2)\lambda_D\big]\frac{d\tau}{N-1} - (N-2)\lambda_D d\tau \\ &= \frac{(N-2)(\rho-\rho_1^*)}{D_N} d\tau. \end{aligned}$$

The expression is the same for every j, so it also equals the NEER response, and all cross rates among A's partners are unchanged. □

Proof of Proposition B.1. From (5) and (1), only tradable prices move, so

$$d\log q_{Aj} = \nu_j + \alpha_T\big(d\log P^{\text{own}}_{T_j} - d\log P^{\text{own}}_{T_A}\big),$$

with

$$d\log P^{\text{own}}_{T_i} = (1-\alpha_D)d\log P_{M_i}.$$

Using (51),

$$d\log q_{AB} = \Big(1 - \frac{mN}{N-1}\Big)\nu_B - \frac{m}{N-1}d\tau.$$

The same calculation for $j = C$ gives (45).

For a general tariff vector,

$$d\log P_{M_A} = \frac{1}{N-1}\sum_j(\nu_j + t_j) = d\nu_A^E + \bar{t},$$

while for a partner that imposes no tariff,

$$d\log P_{M_j} = d\nu_A^E - \frac{N}{N-1}\nu_j.$$

Substitution gives (44). Setting $d\log q_{AC} = 0$ and substituting ν_C from (31) gives

$$\big[(N-1) - mN\big]\frac{\rho - N\rho_1^*}{N} = mD_N(\rho).$$

Solving for ρ gives (46). If $m < 1/N$, the coefficient on the left is positive, implying $\rho_q^*(N) > N\rho_1^*$. Averaging (45) and using (32) gives (47), whose two terms are negative under the stated conditions. □

Proof of Proposition 6.3. The result follows directly from (41). Total differentiation of market clearing gives $dp/dt_B \in (0,1)$, and $M_A' < 0$ and $E_j' > 0$ then imply the stated signs of dq_B/dt_B and dq_C/dt_B. □

Proof of Proposition C.1. Under the weights in (49), the bystanders remain interchangeable, so the system reduces to two unknowns (ν_B, ν_C). The corresponding flow values are $m\beta$, $m\gamma$, and $m\xi$, where

$$\beta = \frac{\kappa}{N-1}, \qquad \gamma = \frac{1-\beta}{N-2}, \qquad \xi = \frac{1-2\gamma}{N-3}.$$

Write

$$\nu_B = \frac{b}{N}, \qquad \nu_C = \frac{c}{N},$$

holding $d\tau$ fixed, and expand the two equilibrium conditions in $1/N$ at fixed (κ, ρ). To leading order,

$$\beta \to \frac{\kappa}{N}, \qquad (N-2)\gamma \to 1,$$

and

$$d\log P_{M_A} = \frac{\kappa\, d\tau + c}{N} + O(N^{-2}), \qquad d\log P_{M_B} = \frac{c-b}{N} + O(N^{-2}), \qquad d\log P_{M_C} = O(N^{-2}).$$

Substituting into (22) and the trade-flow sums gives the limiting system

$$P(b-c) = -\kappa\rho\, d\tau, \qquad P(2c-b) = \kappa(\rho - \rho_1^*)\, d\tau, \qquad P \equiv \rho + \alpha_D(\eta - 1).$$

The limiting matrix

$$P\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

has determinant $P^2 > 0$. Adding the two equations gives

$$c = -\frac{\kappa\rho_1^*}{P} d\tau.$$

Since the NEER weights satisfy $w_{AB} = \beta = O(N^{-1})$ and $(N-2)w_{AC} \to 1$,

$$N\,\frac{d\nu_A^E}{d\tau} \longrightarrow -\frac{\kappa\rho_1^*}{\rho + \alpha_D(\eta-1)}.$$

Finally,

$$\frac{D_N}{N} \longrightarrow \rho + \alpha_D(\eta - 1),$$

which gives (50). $\square$